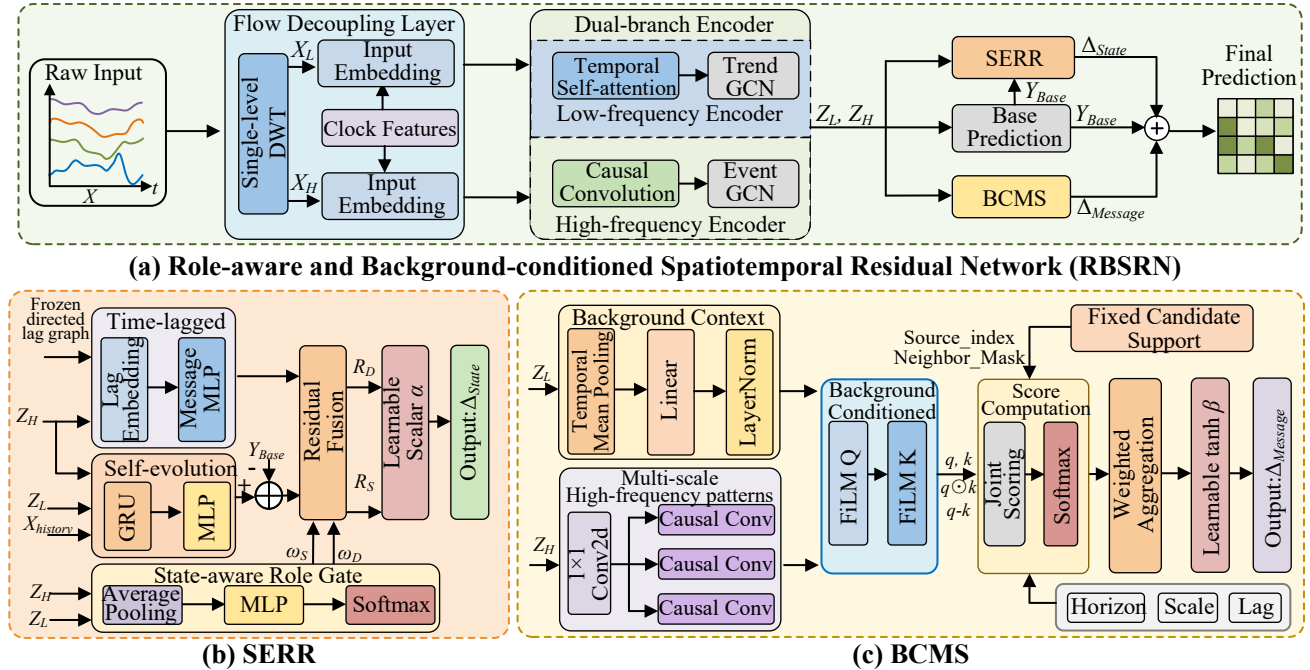

Raw Input
X
t
Flow Decoupling Layer
Single-level DWT
$X_L$
Input Embedding
Clock Features
$X_H$
Input Embedding
Dual-branch Encoder
Temporal Self-attention
Trend GCN
Low-frequency Encoder
Causal Convolution
Event GCN
High-frequency Encoder
$Z_L$, $Z_H$
SERR
$\Delta_{State}$
$Y_{Base}$
Base Prediction
$Y_{Base}$
BCMS
$\Delta_{Message}$
Final Prediction
(a) Role-aware and Background-conditioned Spatiotemporal Residual Network (RBSRN)
Frozen directed lag graph
Time-lagged
Lag Embedding
Message MLP
$Z_H$
Self-evolution
GRU
MLP
$Z_L$
$X_{history}$
$Y_{Base}$
Residual Fusion
$R_D$
$R_S$
Learnable Scalar $\alpha$
Output:$\Delta_{State}$
$\omega_S$
$\omega_D$
State-aware Role Gate
$Z_H$
$Z_L$
Average Pooling
MLP
Softmax
(b) SERR
Background Context
$Z_L$
Temporal Mean Pooling
Linear
LayerNorm
Multi-scale High-frequency patterns
$Z_H$
1×1 Conv2d
Causal Conv
Causal Conv
Causal Conv
Background Conditioned
FiLM Q
FiLM K
Source_index
Neighbor_Mask
Fixed Candidate Support
q, k
$q \odot k$
q-k
Score Computation
Joint Scoring
Softmax
Weighted Aggregation
Learnable tanh $\beta$
Output:$\Delta_{Message}$
Horizon
Scale
Lag
(c) BCMS